

Companion III: Reconstruction, Non-Triviality, and Faithful-Wilson Universality

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This companion contains the detailed proofs for the endpoint part of the theorem-level local-net / vacuum-sector claim. It proves the Euclidean OS structure, Wightman–Haag–Kastler reconstruction, field correspondence, Minkowski-gap transfer, non-triviality, the qualitative faithful-Wilson universality theorem, and the exact bounded-region local-net endpoint. Throughout, the theorem-level object is the local gauge-invariant sharp-local net generated by flowed curvature composites together with its OS/Wightman vacuum representation; within the admissible gradient-flow renormalization class and after coherent normalization, faithful Wilson regularizations yield the same continuum local theory, while the explicit Route 1 quantitative certificate remains ρ -indexed. Extended-support line/surface sectors and other nonlocal sector data remain outside the theorem-level claim. External theorem and appendix references are resolved to the canonical source numbering of the March 16, 2026 manuscript.

GUIDE TO THE DETAILED ARGUMENTS

Appendix M contains the reconstruction and non-triviality material, and Appendix O contains the qualitative faithful-Wilson universality theorem, the ρ -indexed quantitative-status note, and the exact local-net endpoint together with the nonlocal-sector exclusion statement. References that point outside this companion are resolved to the canonical source numbering of the March 16, 2026 manuscript.

Reference orientation. Standard background used across the theorem-level presentation includes Osterwalder–Schrader reconstruction and lattice reflection positivity^{1–3}, Wilson lattice gauge theory and asymptotic freedom^{4–6}, gradient/Wilson flow and heat-kernel tools^{7–9}, cluster expansion and Dobrushin uniqueness^{10–12}, Bessel-ratio and Toeplitz/total-positivity tools used in the one-plaquette comparison arguments^{13–22}, lattice improvement and weak-coupling comparison inputs^{23–25}, the Yang–Mills problem statement²⁶, DHR superselection theory and local quantum physics^{27–31}, and standard real-algebraic geometry background used in auxiliary reductions³².

Cross-reference to detailed proofs.

Core	Src.	Detail proof	First core use
VII.1	M.1	Appendix M (Euclidean OS structure)	Core Theorem VII.1 and then Core Theorem VII.2
VII.2	M.3	Appendix M (OS/Wightman reconstruction)	Core Theorem VII.2 and then Core Corollary VII.3
VII.3	M.4	Appendix M (field-correspondence theorem)	Core Corollary VII.3 and then Core Theorem VII.9
VII.4	M.5	Appendix M (Minkowski-gap transfer)	Core Theorem VII.4 and then Core Theorem I.1(D)
VII.5	M.6	Appendix M (non-triviality theorem)	Core Theorem VII.5 and then Core Theorem I.1(C)
VII.7	O.3	Appendix O (qualitative faithful-Wilson universality)	Core Theorem VII.7 and then Core Corollary VII.8
VII.8	O.4	Appendix O (ρ -indexed quantitative-status note)	Core Corollary VII.8 and then Main Theorem I.1 framing
VII.9	O.5	Appendix O (exact local-net endpoint / sector exclusion)	Core Theorem VII.9 and then Core Corollary VII.10
VII.10	O.7	Appendix O (group-only endpoint form)	Core Corollary VII.10 and then Main Theorem I.1

Appendix M: OS/Wightman theorem package and non-triviality

This appendix is Component 9 in the public theorem decomposition of Section 2. It is invoked directly in Theorem 2.1(C)–(D) and in Appendix O. The constructive inputs remain

in Section 5 (Lane A) and Appendix F (Route 1/QE3), but the point of the present appendix is that a reader can verify the local-field axiomatic output without reverse-engineering the proof spine. For a concise path through the material collected here, one may begin with the theorem stated below and then descend to the companion inputs as needed. We work throughout with the full local gauge-invariant sharp-local state

$$\omega^{(u_{\text{ren}})} \in \mathfrak{A}_{\text{loc}}^*$$

furnished by Corollary 5.76 for a fixed weak-window target value $u_{\text{ren}} \in (0, u_*]$. Write $\mathcal{P}_{+, \text{sharp}}$ for the positive-time sharp-local polynomial algebra from Theorem 5.77. The provenance summary in §M 1 is the front-end guide for this appendix: it separates imported standard OS/Wightman background from earlier internal Euclidean inputs and from the theorem-level endpoints first assembled here.

1. Component 9 provenance summary

For this appendix the provenance bins are used in the following strict sense. *Imported standard background family* means a standard OS/Wightman reconstruction, analytic-continuation, or uniqueness theorem used as a black-box input; *earlier internal theorem* means a theorem already proved before Appendix M; and *new appendix endpoint/reformulation* means a theorem-level output first assembled here from those earlier ingredients.

Component 9 provenance summary. The provenance bins used in this appendix are as follows.

Theorem M.1 (new endpoint)::

consumes Theorem 5.74(b), Corollary 5.76, Theorem F.247, and Theorem 5.77, together with the standard OS reconstruction / uniqueness theorem family for item (v). It gives the full Euclidean OS package on the full sharp-local algebra.

Corollary M.2 (new reformulation)::

consumes Theorem M.1 and Corollary F.5. It converts QE3 into positive-time Euclidean clustering and vacuum uniqueness.

Standard OS reconstruction / uniqueness theorems^{1,2} (imported background)::

used only in Theorem M.1, item (v), and in Theorem M.3.

Theorem M.3 (new endpoint)::

consumes Theorem M.1 together with the imported OS reconstruction / analytic-continuation / uniqueness family. It reconstructs the Wightman fields, spectrum condition, and Haag–Kastler net.

Corollary M.4 (new endpoint)::

consumes Definition F.239, Theorem L.8, Corollary 5.76, and Theorem M.3. It identifies the reconstructed local fields with the Lane A renormalized gauge-invariant local differential polynomials.

Corollary M.5 (new endpoint)::

consumes Theorem M.3 and Corollary F.5. It transfers the OS spectral gap to the Minkowski Hamiltonian.

Theorem M.6 (new endpoint)::

consumes Theorem 5.78, Corollary M.4, Corollary F.5, and Corollary M.5. It rules out the zero/identity theory.

2. Euclidean OS package

Direct inputs. The internal theorem-level input is Theorem 5.74(b), Corollary 5.76, Theorem F.247, and Theorem 5.77; only the standard OS reconstruction / uniqueness theorem family is imported here, and it enters solely in item (v).

Theorem M.1 (Euclidean OS axioms for the full sharp-local theory). *Let $\omega^{(u_{\text{ren}})}$ be the full local gauge-invariant sharp-local state of Corollary 5.76. Then the Schwinger distributions generated by the renormalized local fields of Lane A satisfy the following properties on the full sharp-local algebra $\mathfrak{A}_{\text{loc}}$:*

- (i) *Euclidean invariance under translations and rotations.*
- (ii) *Bosonic symmetry under permutation of insertions.*
- (iii) *Osterwalder–Schrader reflection positivity on the full positive-time sharp-local algebra.*
- (iv) *Regularity/temperedness of the Schwinger distributions.*

(v) OS reconstruction on the full positive-time sharp-local algebra, yielding a Hilbert space

$$(\mathcal{H}_{\text{loc}}, \Omega_{\text{loc}}, H_{\text{loc}}),$$

with $\Omega_{\text{loc}} = [1]$ cyclic for the full sharp-local positive-time algebra. Moreover the bounded positive-time base algebra $\mathcal{A}_+$ embeds densely in $\mathcal{H}_{\text{loc}}$.

Proof. Items (i)–(iii) come from the finite-cap sharp-local OS structure theorem, Theorem 5.74(b), together with the inductive-union passage Corollary 5.76. Concretely, Euclidean invariance, bosonic symmetry, and reflection positivity are proved on each fixed engineering cap and then transported compatibly to the full sharp-local algebra because every test polynomial uses only finitely many generators. The finite-cap operator/domain package used there is isolated in Lemma F.244 and supplies the common invariant core on which the sharp-local field polynomials are realized.

Item (iv) is Theorem F.247, which verifies the regularity and temperedness bounds for the full sharp-local Schwinger family. Once items (i)–(iv) are in place, the standard OS reconstruction theorem applies on the full positive-time sharp-local algebra and yields the Hilbert space $(\mathcal{H}_{\text{loc}}, \Omega_{\text{loc}}, H_{\text{loc}})$. The additional density statement for $\mathcal{A}_+$ is then exactly Theorem 5.77, which identifies the bounded base algebra as dense inside that reconstructed Hilbert space. $\square$

Dependency summary.

Inputs. Theorem 5.74(b), Corollary 5.76, Theorem F.247, and Theorem 5.77

Imported standard background. the standard OS reconstruction / uniqueness theorem family, used only for item (v) after the Euclidean package has been established above

Conclusion at this node. the full sharp-local Schwinger family satisfies the Euclidean OS package on the full sharp-local algebra and yields the reconstructed OS Hilbert space with dense bounded base algebra

Used later in. Corollary M.2 and Theorem M.3

Direct inputs. This corollary consumes only Theorem M.1 and the Route 1 vacuum-gap theorem Corollary F.5.

Corollary M.2 (Positive-time Euclidean clustering and vacuum uniqueness). *In the setting of Theorem M.1, let*

$$F^0 := F - \omega^{(u_{\text{ren}})}(F), \quad G^0 := G - \omega^{(u_{\text{ren}})}(G)$$

for $F, G \in \mathcal{P}_{+, \text{sharp}}$. Let $m_* > 0$ be the Route 1 gap constant from Corollary F.5. Then for every $s \geq 0$,

$$\left| \omega^{(u_{\text{ren}})}(\Theta F^0 \tau_s G^0) \right| \leq \| [F^0] \|_{\mathcal{H}_{\text{loc}}} \| [G^0] \|_{\mathcal{H}_{\text{loc}}} e^{-m_* s}.$$

In particular,

$$\ker(H_{\text{loc}}) = \mathbb{C}\Omega_{\text{loc}},$$

so the Euclidean/OS vacuum is unique up to phase.

Proof. Corollary F.5 gives

$$\text{Spec}(H_{\text{loc}}) \subset \{0\} \cup [m_*, \infty), \quad \ker(H_{\text{loc}}) = \mathbb{C}\Omega_{\text{loc}}.$$

For positive-time sharp-local polynomials, OS reconstruction gives the semigroup identity

$$\omega^{(u_{\text{ren}})}(\Theta F^0 \tau_s G^0) = \langle [F^0], e^{-sH_{\text{loc}}} [G^0] \rangle.$$

Since $[F^0]$ and $[G^0]$ lie in $\Omega_{\text{loc}}^\perp$, the spectral theorem yields

$$\| e^{-sH_{\text{loc}}} \upharpoonright_{\Omega_{\text{loc}}^\perp} \| \leq e^{-m_* s},$$

which gives the stated estimate. The vacuum-uniqueness statement is the second clause of Corollary F.5. $\square$

3. Wightman/Haag–Kastler reconstruction and field correspondence

Direct inputs. The new reconstruction endpoint uses Theorem M.1 as its internal Euclidean package and then invokes only the standard Osterwalder–Schrader analytic-continuation / uniqueness theorem family as imported background.

Theorem M.3 (OS/Wightman theorem package for the local gauge-invariant sharp-local theory). *Let $\omega^{(u_{\text{ren}})}$ be as above. Then the Euclidean Schwinger functions of the full sharp-local theory admit a Minkowski reconstruction with the following properties:*

- (i) *there exists a Hilbert space $\mathcal{H}_{\text{loc}}$, a vacuum vector Ω_{loc} , and a strongly continuous unitary representation U of the Poincaré group whose joint spectrum lies in the closed forward light cone;*

- (ii) *for every renormalized local gauge-invariant field generated by Lane A there is a corresponding operator-valued tempered distribution on Minkowski space whose Wightman distributions are the analytic continuations of the Euclidean Schwinger distributions;*
- (iii) *these fields satisfy locality (spacelike commutativity) and generate the vacuum Haag–Kastler net of the local gauge-invariant sharp-local theory;*
- (iv) *the reconstruction is unique up to unitary equivalence once the Euclidean Schwinger functions are fixed.*

Proof. Theorem M.1 supplies the full Euclidean package on the sharp-local algebra: Euclidean invariance, bosonic symmetry, reflection positivity, temperedness, and the OS-positive positive-time algebra on which reconstruction is to be performed. The standard Osterwalder–Schrader reconstruction and analytic-continuation theorem family therefore applies directly; see^{1,2}. It yields the reconstructed Wightman fields, the strongly continuous Poincaré representation with forward-light-cone spectrum, and spacelike locality after continuation to Minkowski signature.

The Haag–Kastler net is then the vacuum local net generated by the smeared reconstructed fields, and uniqueness up to unitary equivalence is the standard uniqueness clause of the same OS/Wightman reconstruction family once the Euclidean Schwinger functions are fixed. No additional nonstandard reconstruction step is inserted here. $\square$

Dependency summary.

Inputs. Theorem M.1

Imported standard background. the standard Osterwalder–Schrader analytic-continuation / uniqueness theorem family

Conclusion at this node. the Euclidean sharp-local Schwinger family reconstructs uniquely to the Wightman fields and Haag–Kastler net of the local gauge-invariant sharp-local theory

Used later in. Corollary M.4, Corollary M.5, and Theorem O.5

Direct inputs. The field-correspondence statement consumes Definition F.239, Theorem L.8, Corollary 5.76, and the reconstruction theorem Theorem M.3.

Corollary M.4 (Field correspondence for gauge-invariant local differential polynomials).

Let $[P]$ be any exact-dimension quotient class from Definition F.239, and let

$$\mathcal{O}_{[P]}^{\text{ren}}(f; \mu)$$

denote the corresponding renormalized sharp-local field constructed by Lane A. Then in the Wightman reconstruction of Theorem M.3 there exists a unique operator-valued tempered distribution

$$\Phi_{[P]}$$

whose Euclidean Schwinger distributions are the analytic continuations of the mixed Schwinger functions of $\mathcal{O}_{[P]}^{\text{ren}}(f; \mu)$. This correspondence is compatible with finite linear combinations, derivatives, total-derivative/Bianchi/symmetry quotient relations, finite engineering truncations, and the inductive-union passage to the full sharp-local algebra. Consequently the reconstructed local field content is exactly the family of gauge-invariant local differential polynomials in the curvature and its covariant derivatives represented by the Lane A sharp-local algebra.

Proof. Definition F.239 and Theorem L.8 identify the exact-dimension quotient classes before any basis is chosen, and Corollary 5.76 constructs compatible renormalized fields for every finite truncation and hence on the inductive union. Applying Theorem M.3 to the Euclidean Schwinger functions of those renormalized fields gives the corresponding Minkowski operator-valued tempered distributions. Because the quotient relations and truncation compatibilities are already built into the Euclidean theory before reconstruction, the same relations persist after analytic continuation. $\square$

Direct inputs. This gap-transfer corollary consumes only Theorem M.3 and the Route 1 spectral-gap statement Corollary F.5.

Corollary M.5 (Minkowski energy gap from the OS spectral gap). *Let P^0 be the Minkowski Hamiltonian in the Wightman reconstruction of Theorem M.3. Assume the Route 1 vacuum-gap theorem, i.e. Corollary F.5, so that*

$$\text{Spec}(H_{\text{loc}}) \subset \{0\} \cup [m_*, \infty).$$

Then

$$\text{Spec}(P^0) \subset \{0\} \cup [m_*, \infty).$$

In particular, the reconstructed local gauge-invariant sharp-local theory has positive energy, a unique vacuum, and a strictly positive Minkowski mass gap in its vacuum sector.

Proof. Under the Osterwalder–Schrader reconstruction, Euclidean time translations analytically continue to the Minkowski time-translation group. Thus the Minkowski Hamiltonian P^0 coincides with the OS generator H_{loc} on the reconstructed Hilbert space; see Theorem M.3. The spectral inclusion therefore transfers directly from Corollary F.5. $\square$

Direct inputs. The non-triviality package consumes Theorem 5.78, Corollary M.4, Corollary F.5, and Corollary M.5; only the final Källén–Lehmann reformulation uses standard background language.

Theorem M.6 (Non-triviality witness and nonzero spectral weight). *Assume the full sharp-local OS/Wightman construction of Theorem M.3 together with the Route 1 vacuum-gap theorem Corollary F.5. Let $\mathcal{E}_{\text{ren}}$ be the canonical centered CP-even dimension-4 scalar witness from Theorem 5.78, and let $\phi_{*,S}$ be the corresponding positive-time reference test function. Then:*

(i)

$$\|\mathcal{E}_{\text{ren}}(\phi_{*,S})\Omega_{\text{loc}}\|^2 = \omega^{(u_{\text{ren}})}\left(\Theta\mathcal{E}_{\text{ren}}(\phi_{*,S})\mathcal{E}_{\text{ren}}(\phi_{*,S})\right) = 1;$$

(ii) *the centered Euclidean and Minkowski two-point functions of $\mathcal{E}_{\text{ren}}$ are therefore nonzero;*

(iii) *the spectral measure of the nonzero vector $\mathcal{E}_{\text{ren}}(\phi_{*,S})\Omega_{\text{loc}}$ under both H_{loc} and the Minkowski Hamiltonian P^0 is a nonzero finite positive measure supported in $[m_*, \infty)$. Equivalently, the Källén–Lehmann measure of the centered two-point function of the witness field is not the zero measure.*

In particular, the reconstructed local gauge-invariant sharp-local theory is non-trivial in the load-bearing sense required later in the paper: it contains a local gauge-invariant field that is not a multiple of the identity and has nonzero vacuum-created spectral weight above the gap.

Proof. The norm identity in (i) is exactly Theorem 5.78, interpreted in the OS Hilbert space. Because $\mathcal{E}_{\text{ren}}$ is centered, the vector $\mathcal{E}_{\text{ren}}(\phi_{*,S})\Omega_{\text{loc}}$ is orthogonal to the vacuum. Hence Corollary F.5 and Corollary M.5 place its Euclidean and Minkowski spectral measures inside $[m_*, \infty)$. Since the vector has norm one, those measures are nonzero finite positive measures. If the centered Euclidean or Minkowski two-point function were identically zero, the corresponding vacuum-created vector would have zero norm, contradicting (i). The Källén–Lehmann reformulation is the standard spectral representation of the centered Wightman

two-point function of a local field in the vacuum representation, applied here to the witness field furnished by Corollary M.4. Thus the theory is non-trivial in the claimed sense. $\square$

Appendix O: Faithful Wilson representation universality and the exact local-net endpoint

This appendix is Component 10 in the public theorem decomposition of Section 2. It is invoked directly in the preamble of Theorem 2.1 and supplies the final theorem-level local-net result for the paper. Its theorem-level tasks are threefold: first, prove faithful-Wilson universality for the continuum sharp-local theory; second, identify the exact local net attached to a compact connected simple group together with its descent under nonfaithful ultraviolet choices; and third, prove that extended-support line/surface observables lie outside that bounded-region local algebra by localization alone. A separate AQFT commentary remark then records the standard sector interpretation of those excluded observables, and a short closing remark records how the theorem-level local-net endpoint matches the local-field reading of the Yang–Mills existence problem²⁶. The provenance summary in §O 1 is the front-end guide for this appendix: it separates earlier internal universality / descent inputs from the theorem-level endpoint theorems, the imported OS uniqueness clause used in faithful-Wilson universality, and the AQFT commentary that remains explicitly outside the theorem statement.

Claim-hygiene summary. Throughout this appendix, faithful-Wilson universality is qualitative only and is always understood within the admissible gradient-flow renormalization class and after coherent normalization. The explicit Component 1 / Route 1 quantitative certificate remains auxiliary-regularization dependent, and extended-support line / surface sectors are excluded from the theorem-level endpoint by localization rather than by any additional sector analysis.

1. Component 10 provenance summary

For this appendix the provenance bins are used in the following strict sense. *Imported standard background family* means either the standard uniqueness clause of OS reconstruction used in Theorem O.3 or the standard AQFT commentary used only in the closing remarks;

earlier internal theorem means a theorem already proved before Appendix O; and *new appendix endpoint/reformulation* means a theorem-level output first assembled here from those earlier ingredients.

Component 10 provenance summary. The provenance bins used in this appendix are as follows.

Lemma O.1 (new auxiliary lemma)::

consumes only Lemma N.1 and builds the common continuum-normalized faithful-Wilson comparison family.

Theorem O.2 (new endpoint)::

consumes Lemma O.1, Corollary N.21, Proposition F.3, Corollary F.257, and Theorems F.258, F.249, F.268. It closes the uniform faithful-Wilson comparison-family hypotheses.

Theorem O.3 (new endpoint)::

consumes Theorem O.2, Corollary 5.76, Corollary M.4, and the standard OS uniqueness clause. It identifies all faithful Wilson regularizations with one continuum local theory.

Standard OS uniqueness clause^{1,2} (imported background)::

used only in Theorem O.3.

Corollary O.4 (new reformulation)::

consumes Theorem O.3, Corollary N.21, Proposition F.3, and Corollary F.5. It records the faithful- ρ indexing of the explicit Route 1 quantitative certificate.

Theorem O.5 (new endpoint)::

consumes Theorem O.3, Corollary M.4, Corollary 5.76, Lemma 2.5, and Lemma 2.6. It fixes the exact local-net endpoint and separates extended-support data from the theorem claim.

Corollary O.7 (new reformulation)::

consumes Theorem O.3 and Theorem O.5. It restates the endpoint in the group-only language of Theorem 2.1.

Standard AQFT sector interpretation^{27–29} (imported background)::

used only in the closing remarks following Theorem O.5; it is not a theorem-level input to the endpoint statements.

Direct inputs. This appendix-specific comparison lemma uses only the compact-simple local-geometry input Lemma N.1.

Lemma O.1 (Faithful Wilson plaquette actions form a common continuum-normalized action family). *Let G be compact connected simple and let ρ_1, ρ_2 be faithful finite-dimensional unitary Wilson representations. For $i = 1, 2$ define*

$$A_{\rho_i}(g) := 1 - \frac{1}{d_{\rho_i}} \Re \text{Tr}_{\rho_i}(g).$$

Then there exist constants $\kappa_{\rho_i} > 0$ such that in exponential coordinates

$$A_{\rho_i}(e^X) = \frac{\kappa_{\rho_i}}{2} \|X\|^2 + O(\|X\|^4) \quad (\|X\| \rightarrow 0). \quad (\text{O1})$$

Consequently, if one introduces two labeled plaquette slots γ_1, γ_2 supported on the same geometric plaquette, sets

$$V_{\gamma_i} := \kappa_{\rho_i}^{-1} A_{\rho_i}, \quad \mathbf{c}^{(1)} := (1, 0), \quad \mathbf{c}^{(2)} := (0, 1),$$

and lets $\mathcal{C} := \{\mathbf{c}^{(1)}, \mathbf{c}^{(2)}\} \subset [0, \infty)^2$, then the two faithful Wilson lattice actions are both members of one continuum-normalized action family in the sense of Definition F.253.

Proof. By Lemma N.1, the quadratic Taylor coefficient of $A_{\rho_i}(e^X)$ is a positive definite Ad-invariant quadratic form, hence equals $\frac{\kappa_{\rho_i}}{2} \|X\|^2$ for some $\kappa_{\rho_i} > 0$. Because $d_{\rho_i}(X)$ is skew-Hermitian, every odd power $d_{\rho_i}(X)^{2m+1}$ is again skew-Hermitian and therefore has purely imaginary trace. Taking the real part of the exponential series for $\text{Tr}_{\rho_i}(e^X)$ removes all odd terms, so the first omitted term after the quadratic contribution is quartic. This proves (O1). Now choose the labeled plaquette slots and coefficient vectors exactly as stated. Each V_{γ_i} has quadratic coefficient 1, and for each coefficient vector one has $\sum_{\gamma} c_{\gamma} \kappa_{\gamma} = 1$. Therefore both faithful Wilson actions belong to the same continuum-normalized compact action family. $\square$

Direct inputs. The uniform faithful-Wilson hypothesis theorem consumes Lemma O.1, Corollary N.21, Proposition F.3, Corollary F.257, and the action-family universality theorems cited in its statement.

Theorem O.2 (Faithful Wilson comparison families satisfy the Route 1 universality hypotheses uniformly). *Let G be compact connected simple and let ρ_1, ρ_2 be faithful finite-dimensional unitary Wilson representations. Form the finite continuum-normalized action family $\mathcal{C} = \{\mathbf{c}^{(1)}, \mathbf{c}^{(2)}\}$ from Lemma O.1, and impose the same admissible gradient-flow renormalization prescription of Definition F.269. Then the non-collapse / one-shot entrance and UV-matching hypotheses required by Theorem F.258 hold uniformly on this faithful-Wilson comparison family. Equivalently, the uniform entrance/non-collapse hypothesis invoked by Theorems F.249 and F.268 is satisfied for this family.*

More concretely, for each $i = 1, 2$ let

$$\beta_*^{(i)} := \beta_*(G, \rho_i), \quad q_*^{(i)} := q_*(G, \rho_i), \quad C_{\text{ent}}^{(i)} := C_{\text{ent}}(G, \rho_i),$$

and let $N_{\text{ent}}^{(i)}(u_{\text{ren}})$ be the tuned-sequence entry index attached to the faithful regularization ρ_i by Proposition F.3. Then the uniform choices

$$\beta_*^{\text{fam}} := \max\{\beta_*^{(1)}, \beta_*^{(2)}\}, \quad q_*^{\text{fam}} := \max\{q_*^{(1)}, q_*^{(2)}\} < 1,$$

$$C_{\text{ent}}^{\text{fam}} := \max\{C_{\text{ent}}^{(1)}, C_{\text{ent}}^{(2)}\}, \quad N_{\text{ent}}^{\text{fam}}(u_{\text{ren}}) := \max\{N_{\text{ent}}^{(1)}(u_{\text{ren}}), N_{\text{ent}}^{(2)}(u_{\text{ren}})\}$$

work simultaneously for both faithful regularizations, each at its own one-shot scale $B(\beta; G, \rho_i)$. Consequently the faithful-Wilson comparison family meets the hypotheses of Theorems F.258, F.249, and F.268.

Proof. By Corollary N.21 and Proposition F.3, each faithful regularization ρ_i separately satisfies the one-shot entrance and bounded-physical-entrance package with explicit constants

$$\beta_*(G, \rho_i), \quad q_*(G, \rho_i), \quad C_{\text{ent}}(G, \rho_i),$$

and with its own entrance block scale $B(\beta; G, \rho_i)$. Because the comparison family is finite, taking maxima over $i = 1, 2$ yields common constants β_*^{fam} , $C_{\text{ent}}^{\text{fam}}$, and $N_{\text{ent}}^{\text{fam}}(u_{\text{ren}})$, while $q_*^{\text{fam}} < 1$ remains a strict contraction constant. Thus the non-collapse / one-shot entrance hypothesis required in Theorem F.258 is uniform across the two faithful Wilson regularizations.

For the UV-matching input, Corollary F.257 already proves the required entrance-scale matching estimate uniformly on any compact continuum-normalized action family. Here the coefficient set $\mathcal{C} = \{\mathbf{c}^{(1)}, \mathbf{c}^{(2)}\}$ is finite, hence compact, so the same UV-matching estimate applies uniformly to the faithful-Wilson comparison family. This closes the hypotheses

of Theorem F.258. Since Theorems F.249 and F.268 invoke exactly that same uniform entrance/non-collapse package for the action class, they apply as well. $\square$

Direct inputs. The faithful- ρ universality theorem consumes Theorem O.2, Corollary 5.76, Corollary M.4, and only the standard uniqueness clause of OS reconstruction as imported background.

Theorem O.3 (Qualitative faithful Wilson representation universality). *Let G be compact connected simple and let ρ_1, ρ_2 be faithful finite-dimensional unitary Wilson representations. Construct the corresponding continuum theories using the same admissible gradient-flow renormalization scheme of Definition F.269 and the same coherent normalization conditions of Lemma F.267. Then there exists a unique $*$ -algebra isomorphism*

$$\Phi_{\rho_1 \rightarrow \rho_2} : \mathfrak{A}_{\text{loc}}(G, \rho_1) \xrightarrow{\cong} \mathfrak{A}_{\text{loc}}(G, \rho_2)$$

which, on every finite engineering cap, sends the Lane A renormalized field attached to an exact-dimension quotient label $[P]$ and test function f to the renormalized field with the same label $[P]$ and the same test function f . The induced map intertwines all finite-family Schwinger functions, the regionwise local nets, the OS vacuum reconstructions, and the reconstructed Wightman fields. In particular, within the admissible gradient-flow renormalization class and after coherent normalization, the continuum local gauge-invariant sharp-local Yang–Mills theory and its vacuum gap depend only on G and not on the faithful Wilson representation used to regularize the ultraviolet lattice theory.

Proof. By Lemma O.1, the two faithful Wilson plaquette actions belong to one continuum-normalized compact action family. For that family we impose the same admissible flow-based renormalization prescription and the same coherent normalization conditions. Theorem O.2 verifies that this finite faithful-Wilson comparison family satisfies the uniform Route 1 entrance/non-collapse and UV-matching hypotheses required by Theorems F.258, F.249, and F.268. Hence the finite-family universality theorem for the renormalized local field algebra and its inductive-union upgrade apply simultaneously to the two faithful Wilson regularizations.

Fix a finite engineering cap Δ_{max} and a finite family of exact-dimension quotient labels $[P_1], \dots, [P_m]$ with test functions $f_1, \dots, f_m$. The cited universality theorems imply that the

limiting Schwinger functions of the corresponding Lane A renormalized fields agree for the two faithful regularizations:

$$\omega_{\rho_1}\left(\prod_{r=1}^m \mathcal{O}_{[P_r]}^{\text{ren},\rho_1}(f_r)\right) = \omega_{\rho_2}\left(\prod_{r=1}^m \mathcal{O}_{[P_r]}^{\text{ren},\rho_2}(f_r)\right).$$

Hence on that finite cap the assignment

$$\mathcal{O}_{[P]}^{\text{ren},\rho_1}(f) \longmapsto \mathcal{O}_{[P]}^{\text{ren},\rho_2}(f)$$

preserves all capwise field relations and all mixed moments. Corollary 5.76 makes these capwise identifications compatible on overlaps of engineering caps, so they glue to a unique isomorphism of the inductive-union sharp-local algebra.

Corollary M.4 identifies both reconstructed Minkowski field families with the same gauge-invariant local differential polynomials in the curvature and its covariant derivatives, so this isomorphism is canonical at the theorem level. Because the full Schwinger families agree under this identification, the standard uniqueness clause of Osterwalder–Schrader reconstruction gives a unitary equivalence of the corresponding OS vacuum reconstructions and hence of the Wightman theories reconstructed in Appendix M. The vacuum Hamiltonians are therefore unitarily equivalent and have the same spectral gap; this is a qualitative universality theorem, not a representation-uniform quantitative comparison of the Route 1 constants. $\square$

Dependency summary.

Inputs. Theorem O.2, Corollary 5.76, and Corollary M.4

Imported standard background. the standard uniqueness clause of Osterwalder–Schrader reconstruction

Conclusion at this node. all faithful Wilson regularizations of the same compact connected simple group determine one canonically identified continuum sharp-local local theory; the theorem does not claim representation-uniform quantitative Route 1 constants

Used later in. Corollary O.4, Theorem O.5, and Corollary O.7

Direct inputs. This quantitative-status corollary consumes Theorem O.3, Corollary N.21, Proposition F.3, and Corollary F.5.

Corollary O.4 (The explicit Route 1 certificate remains indexed by the auxiliary faithful regularization). *Fix a compact connected simple group G and choose a faithful finite-dimensional unitary Wilson regularization ρ . Theorem O.3 identifies all faithful choices, within the*

admissible gradient-flow renormalization class and after coherent normalization, with the same continuum local theory

$$\mathfrak{A}_{\text{loc}}(G),$$

but the explicit Component 1 / Route 1 certificate retained in this manuscript is currently recorded through the ρ -indexed constants

$$\beta_*(G, \rho), \quad q_*(G, \rho), \quad C_{\text{ent}}(G, \rho), \quad m_{\text{blk}}(G, \rho), \quad \mathfrak{m}_*(G, \rho; u_{\text{ren}}).$$

The manuscript does not claim that these quantitative constants are uniform over all faithful ρ . Different faithful Wilson regularizations yield canonically isomorphic continuum local theories, but may certify different explicit lower bounds on their common vacuum gap.

Proof. Theorem O.3 identifies the continuum local theory for any two faithful regularizations. On the other hand, the explicit constants listed above are read from the chosen faithful ultraviolet regularization through Corollary N.21, Proposition F.3, and Corollary F.5. No representation-uniform quantitative comparison theorem for those explicit constants is proved in the present manuscript. $\square$

Direct inputs. The exact endpoint theorem consumes Theorem O.3, Corollary M.4, Corollary 5.76, Lemma 2.5, and Lemma 2.6; the AQFT sector interpretation enters only afterward in the commentary remark.

Theorem O.5 (Exact local-net endpoint and exclusion of extended-support sector data). *Fix a compact connected simple gauge group G . Choose any faithful finite-dimensional unitary Wilson representation ρ_{fid} of G and let*

$$\mathfrak{A}_{\text{loc}}(G, \rho_{\text{fid}}), \quad (\mathcal{H}_{\text{loc}}(G, \rho_{\text{fid}}), \Omega_{\text{loc}}(G, \rho_{\text{fid}}), U_{G, \rho_{\text{fid}}})$$

be the sharp-local Euclidean/Wightman theory produced from that faithful ultraviolet regularization by Corollary 5.76 and Appendix M. By Theorem O.3, different faithful choices of ρ_{fid} yield canonically isomorphic theories, so we may write

$$\mathfrak{A}_{\text{loc}}(G), \quad (\mathcal{H}_{\text{loc}}(G), \Omega_{\text{loc}}(G), U_G)$$

for that canonical faithful- ρ -independent local theory, always understood within the admissible gradient-flow renormalization class and after coherent normalization. Then:

(i) by Corollary M.4, the local Wightman fields in this theory are precisely the renormalized gauge-invariant local differential polynomials in the curvature and its covariant derivatives represented by Lane A;

(ii) if ρ is any finite-dimensional unitary Wilson representation of G and $G_{\text{eff}} := G / \ker(\rho)$, then the local observable net obtained from the ultraviolet pair (G, ρ) is canonically identified with

$$\mathfrak{A}_{\text{loc}}(G_{\text{eff}}).$$

In particular, for faithful ρ this is just the canonical theory $\mathfrak{A}_{\text{loc}}(G)$ of Theorem O.3;

(iii) line operators, surface defects, and other global-form-sensitive observables are not elements of the bounded-region local field algebra constructed in Appendix M, because their localization regions are extended rather than bounded.

Consequently the theorem-level object constructed in this manuscript is, within the admissible gradient-flow renormalization class and after coherent normalization, the canonical local gauge-invariant sharp-local net $\mathfrak{A}_{\text{loc}}(G)$ together with its OS/Wightman vacuum representation, and no extended-support nonlocal sector data enter that theorem-level existence-and-gap claim.

Proof. Item (i) is Corollary M.4, which identifies the reconstructed Minkowski fields with the renormalized gauge-invariant local differential polynomials represented by the Lane A sharp-local algebra. For item (ii), if ρ is faithful then the claim is exactly Theorem O.3. If ρ is not faithful, combine the descent-to- G_{eff} statement for the Wilson action and Schwinger functions (Lemma 2.5) with the adjoint-local generator theorem (Lemma 2.6) and then apply Theorem O.3 to the descended faithful representation on G_{eff} .

For item (iii), Appendix M constructs operator-valued tempered distributions localized in bounded spacetime regions. A Wilson line, 't Hooft line, or higher-dimensional defect depends on extended geometric support and is therefore outside that bounded-region local field algebra by localization alone. Thus the theorem-level object stops at the local net and its vacuum representation; any AQFT discussion of sectors built from extended-support observables is commentary downstream of this theorem, not part of its proof burden. $\square$

Dependency summary.

Inputs. Theorem O.3, Corollary M.4, Corollary 5.76, Lemma 2.5, and Lemma 2.6

Imported standard background. none; the AQFT sector interpretation is commentary only and enters only in the following remark

Conclusion at this node. the theorem-level object is the canonical bounded-region local gauge-invariant sharp-local net together with its OS/Wightman vacuum representation; extended-support observables are excluded from that theorem object by localization alone

Used later in. Corollary O.7 and the final framing of Theorem 2.1

Remark O.6 (AQFT commentary on the excluded extended-support observables). Theorem O.5(iii) stops at a theorem-level localization statement: the bounded-region local field algebra constructed in Appendix M does not contain observables whose geometric support is extended. In the standard algebraic-QFT description, such charged or nonlocal objects are then organized as sectors, representations, or field-algebra extensions over the same local observable net rather than as elements of the local observable net itself^{27–29}.

Direct inputs. This main-theorem restatement consumes only Theorem O.3 and Theorem O.5.

Corollary O.7 (Main-theorem group-only local-net endpoint and non-claims). *Theorem 2.1 is a theorem about the local gauge-invariant Yang–Mills theory attached to each compact connected simple gauge group G . Any faithful finite-dimensional unitary Wilson representation may be used as an auxiliary ultraviolet regularization and, within the admissible gradient-flow renormalization class and after coherent normalization, yields the same continuum local theory*

$$\mathfrak{A}_{\text{loc}}(G).$$

The explicit Route 1 certificate constants retained in Theorem 2.1 remain indexed by the chosen faithful ultraviolet regularization, as recorded in Corollary O.4. A nonfaithful ultraviolet choice yields instead the local theory of the effective quotient $G_{\text{eff}} := G/\ker(\rho)$. Differences between global forms survive only in nonlocal sector data not asserted by Theorem 2.1; differences between faithful ultraviolet Wilson representations survive only through the present quantitative certificate, not through the continuum local theory itself.

Proof. This is immediate from Theorems O.3 and O.5. □

Remark O.8 (How Component 10 matches the local-field formulation). The official local-field formulation of the Yang–Mills existence problem asks for local quantum fields corresponding to the gauge-invariant local polynomials in the curvature F and its covariant derivatives,

together with axiomatic properties at least as strong as the OS/Wightman frameworks²⁶. Theorem O.5(i) identifies exactly those local fields inside the canonical net $\mathfrak{A}_{\text{loc}}(G)$, while Corollary O.7 records that faithful Wilson regularizations produce that same local theory and that nonfaithful ultraviolet choices descend to the effective quotient. The theorem-level endpoint is therefore the local gauge-invariant Yang–Mills theory generated by flowed curvature composites, whereas extended-support line/surface observables remain downstream sector data over that same local net.

Remark O.9 (Further sector completions lie downstream of the public theorem). The manuscript does not deny the mathematical interest of the global-form-sensitive sector problem. On the contrary, once the local net has been constructed with OS/Wightman strength and a vacuum mass gap, it is natural to ask for a sharper analysis of line/surface operators, charged sectors, and possible field-algebra completions. The point is only logical: those questions are downstream extensions of the local existence-and-gap theorem, not part of the theorem-level object already proved here.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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